

Multi-Agent Strategic Reasoning for the Imperfect Information Game Avalon

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1 Project Overview

This project studies how large language models perform strategic reasoning and social deduction in an imperfect-information multi-agent environment. Using *Avalon* as a controlled testbed, I implemented a full Python simulation of the game, covering role assignment, private information flow, mission proposal and voting, hidden success/fail decisions, and the endgame assassination phase. The engine records structured trajectories—including team proposals, public votes, mission outcomes, and multi-turn dialogue—providing a reliable basis for decision-level and communication-level analysis.

On top of this environment, I instantiate six heterogeneous dialogue agents, each conditioned on a different conversational corpus and persona profile. These agents exhibit distinct linguistic styles, risk tolerance, and argumentation behavior, allowing the study of how dataset-driven social priors interact with hidden-role inference and deception dynamics.

Central to the system is a recursive reasoning module implemented in the `Player` class. For each decision point, an agent: (1) interprets its private role, (2) extracts evidence from dialogue, voting patterns, and mission history, (3) performs multi-stage memory filtering combining short-term, long-term, and semantic-retrieval memory, and (4) updates probabilistic beliefs about each player’s alignment. These beliefs guide the agent’s next utterance or action, maintaining internal consistency with its persona and strategic objectives.

Together, the environment, persona-grounded agents, and recursive reasoning process form a unified platform for examining how LLMs reason, infer, and communicate under uncertainty in cooperative–adversarial settings.

2 Python Game Logic

To provide a controlled testbed for multi-agent social deduction, we implemented a complete, rule-faithful *Avalon* engine in Python. The environment follows the canonical structure of the game: hidden role assignment, sequential mission rounds, public discussion and voting, private success/fail actions, and the final assassination stage. This implementation enables systematic analysis of both decision-making and dialogue-based inference under imperfect information.

- **Role initialization and private information.** At the start of each episode, roles such as Merlin, Percival, Morgana, Assassin, Oberon, and loyal/evil servants are sampled and assigned to `Player` instances. Each agent receives role-specific descriptions and privileged knowledge when appropriate (e.g., evil players knowing one another, Merlin knowing all evil roles). This initializes heterogeneous prior beliefs that guide downstream reasoning.
- **Round structure and mission mechanics.** Each round proceeds through Avalon’s standard sequence: a leader proposes a mission team; all agents cast public approval votes; approved teams privately submit success/fail decisions; and the aggregated mission result is revealed. The engine stores team proposals, vote distributions, and hidden actions, allowing agents to perform history-conditioned inference across multiple rounds.
- **Dialogue integration and memory architecture.** During discussion phases, agents generate natural-language statements via the `Player` module. The engine logs these utterances, and each agent maintains structured memory buffers including short-term conversational memory, long-term summary memory, rule memory, and a dynamic core memory used for role inference. A

hybrid filtering mechanism—combining recency weighting, informativeness scoring, and semantic similarity via Sentence-BERT—keeps memory compact while retaining conceptually useful evidence.

- **Hidden actions and endgame resolution.** Mission outcomes are computed while preserving anonymity of individual actions. Standard Avalon win conditions are implemented: the good team succeeds with three clean missions unless the Assassin correctly identifies Merlin during the final assassination phase. All intermediate states, including dialogue and belief updates, are preserved for downstream inspection.
- **Trajectory generation for multi-agent reasoning.** The engine outputs complete symbolic and linguistic trajectories containing role assignments, proposals, votes, mission outcomes, dialogue turns, and per-agent belief updates. These trajectories serve as structured supervision for analyzing the alignment between conversational behavior, hidden-role inference, and strategic decision-making.

3 Personality Corpora and the Six Agents

Each of the six agents is designed as a distinct conversational and strategic archetype, created by combining a fixed persona prompt with one primary dialogue corpus. These corpora introduce different linguistic priors—such as self-consistency, emotional sensitivity, topical grounding, and task-oriented planning—allowing us to systematically study how dataset-induced styles influence hidden-role inference and deception in social deduction.

- **Persona-consistent conversationalist (Persona-Chat / ConvAI2)** [ZDU⁺18]. Drawn from persona-grounded dialogues in which speakers must remain faithful to a predefined textual identity. This agent exhibits coherent self-description, stable tone, and a grounded personal narrative throughout the game.
- **Skill-blending social player (Blended Skill Talk)** [SWS⁺20]. Trained on conversations blending personality, empathy, and external knowledge, producing an agent capable of fluidly shifting between supportive dialogue, persuasion, and cooperative reasoning during proposals and voting.
- **Everyday casual speaker (DailyDialog)** [LSS⁺17]. Based on clean, human-written everyday conversations annotated with communicative intentions and emotions. The corresponding agent adopts relaxed, informal discourse patterns resembling casual human gameplay.
- **Emotion-aware empath (EmpatheticDialogues)** [RHH⁺19]. Derived from emotionally grounded conversations, enabling the agent to generate empathic responses, de-escalate conflict, and interpret suspicion or frustration through an affectively sensitive lens.
- **Structured planner (MultiWOZ 2.2)** [ZRS⁺20]. A large task-oriented dialogue corpus with explicit dialogue-state tracking. This agent tends to reason in terms of constraints, plans, and long-term objectives, exhibiting structured argumentation when evaluating mission teams.
- **Topical debater (Topical-Chat)** [GHC⁺19]. A knowledge-grounded conversation dataset supporting discussions over diverse topics. The resulting agent frequently justifies suspicions with broader context and uses abstract, topic-driven reasoning during debate.

Together, these six corpora anchor qualitatively distinct player types, enabling controlled experimentation on how conversational style and linguistic priors affect role inference, deception, cooperation, and persuasion in imperfect-information multi-agent settings.

4 Recursive Reasoning Pipeline

To model iterative human-like deliberation in a social deduction setting, each agent executes a multi-stage recursive reasoning routine at every decision point (discussion, team proposal, voting, and endgame assassination). This procedure integrates private-role understanding, structured memory retrieval, and belief updating to support coherent long-horizon strategy formation.

- **Private role interpretation.** At the start of each round, the agent retrieves a role-specific rule description detailing its win condition, visibility privileges, and interaction constraints. Evil-aligned roles receive shared information about teammates, while Merlin and Percival obtain role-specific partial information. This initializes the agent’s internal prior over player alignments.
- **Evidence-driven identity inference.** The agent assembles evidence from multiple memory buffers—including short-term conversational memory, long-term summaries, mission history, and a dedicated rule memory. Dialogue, votes, and mission outcomes are jointly analyzed to extract signals about which players are likely good or evil, identifying specific utterances or events that contradict or reinforce role expectations.
- **Two-stage memory filtering.** A coarse retrieval pass selects broad candidate memories using recency and interaction frequency. This is followed by a fine-grained semantic filtering stage using Sentence-BERT similarity scoring, which highlights conceptually informative statements (e.g., inconsistent defenses, suspicious voting patterns) and suppresses redundant or noisy events. The result is a compact working memory serving as the agent’s contextual evidence set.
- **Belief update and strategy adaptation.** The agent synthesizes its filtered evidence into a probabilistic belief model over all players’ alignments. This belief state conditions its next action: supporting or rejecting mission teams, generating accusations or defenses, or forming persuasive arguments consistent with its persona. Agents update beliefs recursively after each utterance or mission outcome, enabling non-myopic, round-to-round adaptation.

This recursive pipeline allows agents to produce strategically coherent, personality-aligned dialogue and decisions, yielding behavior that is more interpretable and stable than single-shot prompting. By combining structured memory management, semantic retrieval, and probabilistic belief formation, the system captures key aspects of human deduction and adaptation in imperfect-information multi-agent games.

5 Observations and Outlook

Experiments with the six heterogeneous agents show that linguistic priors inherited from different conversational corpora exert a measurable influence on strategic behavior in *Avalon*. Agents conditioned on persona-grounded or knowledge-intensive datasets tend to generate more assertive and structured arguments, whereas those trained on empathetic or everyday dialogue corpora adopt softer, more cautious styles of persuasion. These stylistic differences persist even when agents observe identical game states, indicating that conversational datasets shape not only surface-level phrasing but also deeper preferences in risk-taking, accusation, and defense.

The recursive reasoning pipeline further enhances the agents’ ability to perform identity inference across multiple rounds. By repeatedly integrating private-role information, retrieved dialogue evidence, and mission outcomes, agents develop more consistent long-horizon strategies than those produced by single-shot prompting. This iterative process allows belief states to evolve in response to subtle conversational cues, often leading to coherent patterns of suspicion or trust that align with the underlying game structure.

Notably, several agents demonstrate emergent persuasive or trustworthy behavior, particularly in settings where their persona aligns with cooperative rhetoric. Although these behaviors do not guarantee optimal play, they highlight the complex interplay between linguistic priors, belief modeling, and strategic adaptation. Taken together, these observations suggest that combining persona-grounded dialogue models with structured multi-step reasoning provides a promising framework for studying LLM behavior in social deduction, deception, and other cooperative–adversarial multi-agent environments.

References

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